BOARDREPLAY / LESSON NOTES

Completing the square

Mathematics · Synthetic sample

> SAMPLE DATA · Synthetic board frames and prebuilt notes; no live model output.

A synthetic mathematics lesson showing the original setup, a positive-root-only draft, its visible correction, and verification of both solutions. This is prebuilt SAMPLE DATA, not a live AI response.

> Reconstruction of captured images; not an exact recording of the lecture. Uncaptured erased content cannot be recovered. Derived explanations and user corrections are labeled.

01 · Complete the square

1. OBSERVED · Sources: Photo 1

The original quadratic equation

$$x^2 - 6x + 5 = 0$$

Transcribed sample; symbolic reasoning was not independently verified.

2. OBSERVED · Sources: Photo 1, Photo 2

Move the constant, then add 9 to both sides.

$$\begin{aligned} x^2 - 6x &= -5 \\ x^2 - 6x + 9 &= -5 + 9 \\ x^2 - 6x + 9 &= 4 \end{aligned}$$

Transcribed sample; symbolic reasoning was not independently verified.

3. OBSERVED · Sources: Photo 2, Photo 3

Recognize the perfect square.

$$(x - 3)^2 = 4$$

Transcribed sample; symbolic reasoning was not independently verified.

02 · Preserve the correction

4. OBSERVED · Sources: Photo 2, Photo 3

Earlier draft: only the positive square root was written. This incomplete version remains in the timeline.

$$x - 3 = 2$$

Transcribed sample; symbolic reasoning was not independently verified.

> Review: Incomplete draft; corrected in the next captured frame.

5. OBSERVED · Sources: Photo 3

The next frame crosses out the draft and explicitly takes both square roots.

$$\begin{aligned} x - 3 &= \pm 2 \\ x &= 3 \pm 2 \end{aligned}$$

Transcribed sample; symbolic reasoning was not independently verified.

6. DERIVED · Educational explanation · Sources: Photo 2, Photo 3

Taking a square root requires both signs here: two real numbers, 2 and −2, square to 4. This explanation is supplied by the sample author.

Transcribed sample; symbolic reasoning was not independently verified.

03 · Check the result

7. OBSERVED · Sources: Photo 4

The two solutions shown on the final board.

$$\boxed{x = 1 \quad \text{or} \quad x = 5}$$

Transcribed sample; symbolic reasoning was not independently verified.

8. OBSERVED · Sources: Photo 4

The final frame substitutes each root into the original equation.

$$\begin{aligned} 1^2 - 6(1) + 5 &= 0 \\ 5^2 - 6(5) + 5 &= 0 \end{aligned}$$

Direct arithmetic: 1 − 6 + 5 = 0 and 25 − 30 + 5 = 0. This checks these substitutions only.

04 · What remains incomplete

9. OBSERVED · Sources: Photo 4

A separate problem begins on the last board. Its solution was not captured.

$$x^2 + 4x + 4 = 0$$

Transcribed sample; symbolic reasoning was not independently verified.

10. UNCERTAIN · Sources: Photo 1

A homework page number is unreadable in the first frame.

Transcribed sample; symbolic reasoning was not independently verified.

> Review: The page number cannot be determined from the captured image.

Unresolved information

- The homework page number is unreadable.
- The next problem is visible but its solution is absent. Nothing erased before the first photo can be recovered.